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SDEV120 Computing Logic
Ivy Tech Community College
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Jeopardy Game Flowcharts

Module 5 Exercise 1 Flowcharting

OVERVIEW

These flowcharts follow the Jeopardy plan from Exercise 1. The goal was to use only structured logic and to break the game into modules so no group of steps gets repeated. The charts cover picking a clue, buzzing in, checking that responses are phrased as a question, scoring, Final Jeopardy, and announcing the winner. Everything was built in draw.io.

HOW THE CHARTS FIT TOGETHER

main() is the top level chart. Every green box with bars on the sides is a call to a module, and each module has its own chart on the pages after this one. Each module chart starts with its name and ends with *return*, which sends control back to the step right after the call.

Chart	What it does	Structures to look for
main()	Declares the game data, sets up the game, and loops while cluesLeft > 0, calling selectClue() and playClue() each pass. Then it runs Final Jeopardy and shows the results.	Module calls, one main loop
getPlayers()	Gets the number of players (1 to 3) and each name, and sets each score to 0.	Input check loop, counted loop
loadClues()	Reads 5 clues and responses for each of the 6 categories and marks them unused.	Loop inside a loop
selectClue()	Repeats until the player picks a clue that is on the board and has not been played.	Loop with a flag, nested selection
playClue()	Reads the clue and keeps it open while players can still buzz in.	Loop with two conditions
takeTurn()	Gets one response, calls checkResponse(), and adds or subtracts the clue value.	Sequence, selection
checkResponse()	Returns "Y" only if the response starts with question words and the rest matches.	Nested selection
getWagers()	Gets a wager from each player above $0 and checks that it is between $0 and their score.	Selection with a loop inside
finalJeopardy()	Shows the clue, collects each response, and applies the wagers.	Loop, nested selection

Chart	What it does	Structures to look for
showResults()	Shows every score, tracks the best one, and flags a tie.	Loop, nested selection

STRUCTURES USED

Every chart sticks to the three basic structures, and each structure has one entry and one exit.

Sequence

Steps run one after another, top to bottom.

Selection

A diamond tests a condition and takes exactly one Yes or No path. Both paths come back together at a single point (the small black dot).

Loop

The test sits at the top. While it is true, the body repeats and control returns to the same test. When it is false, the No path leaves the loop.

SYMBOL KEY

Start, stop, return	Process	Module call	Input/output	Decision	Rejoin dot

FLOWCHARTS

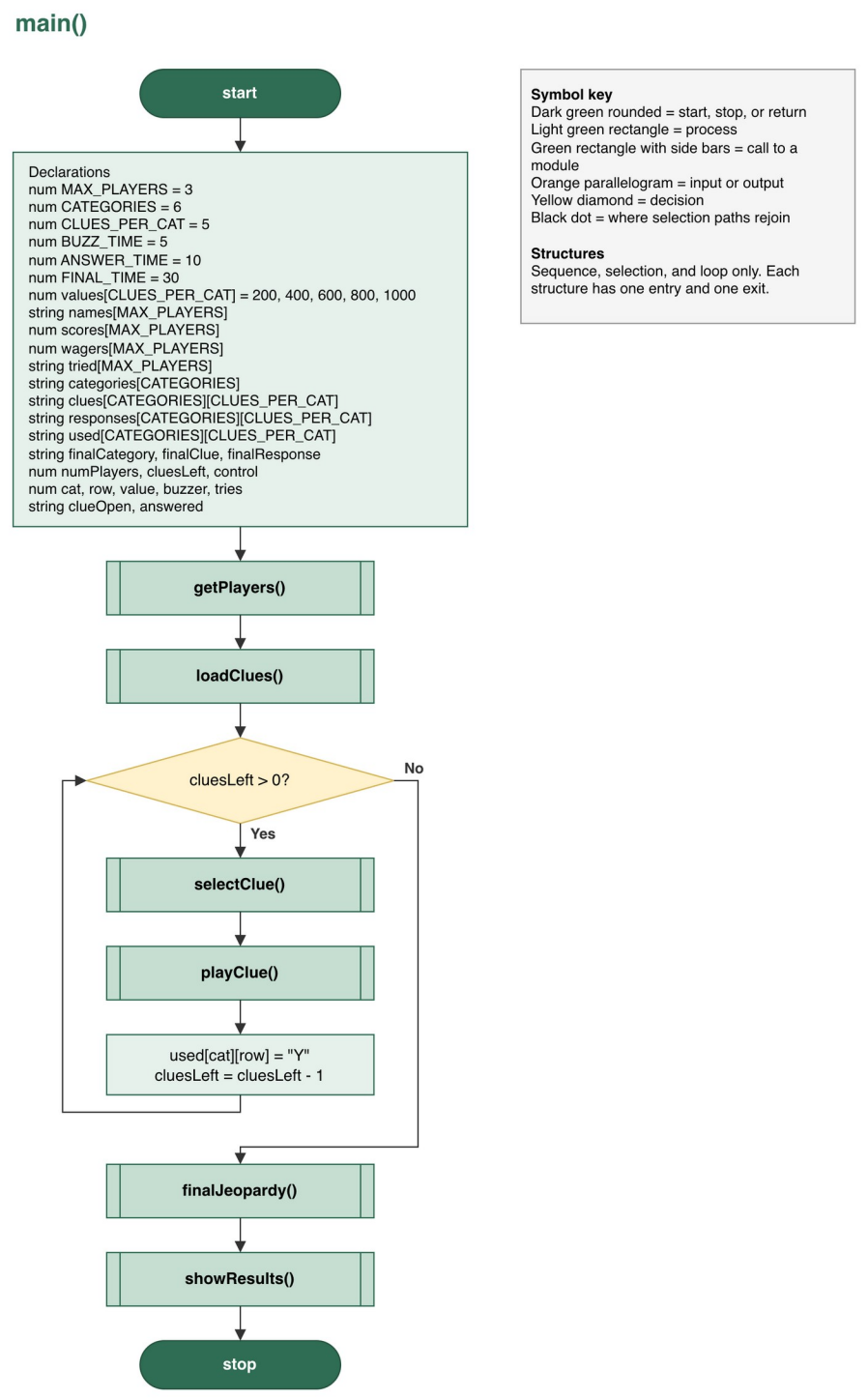


Figure 1 | main() is the top level of the program and calls every other module

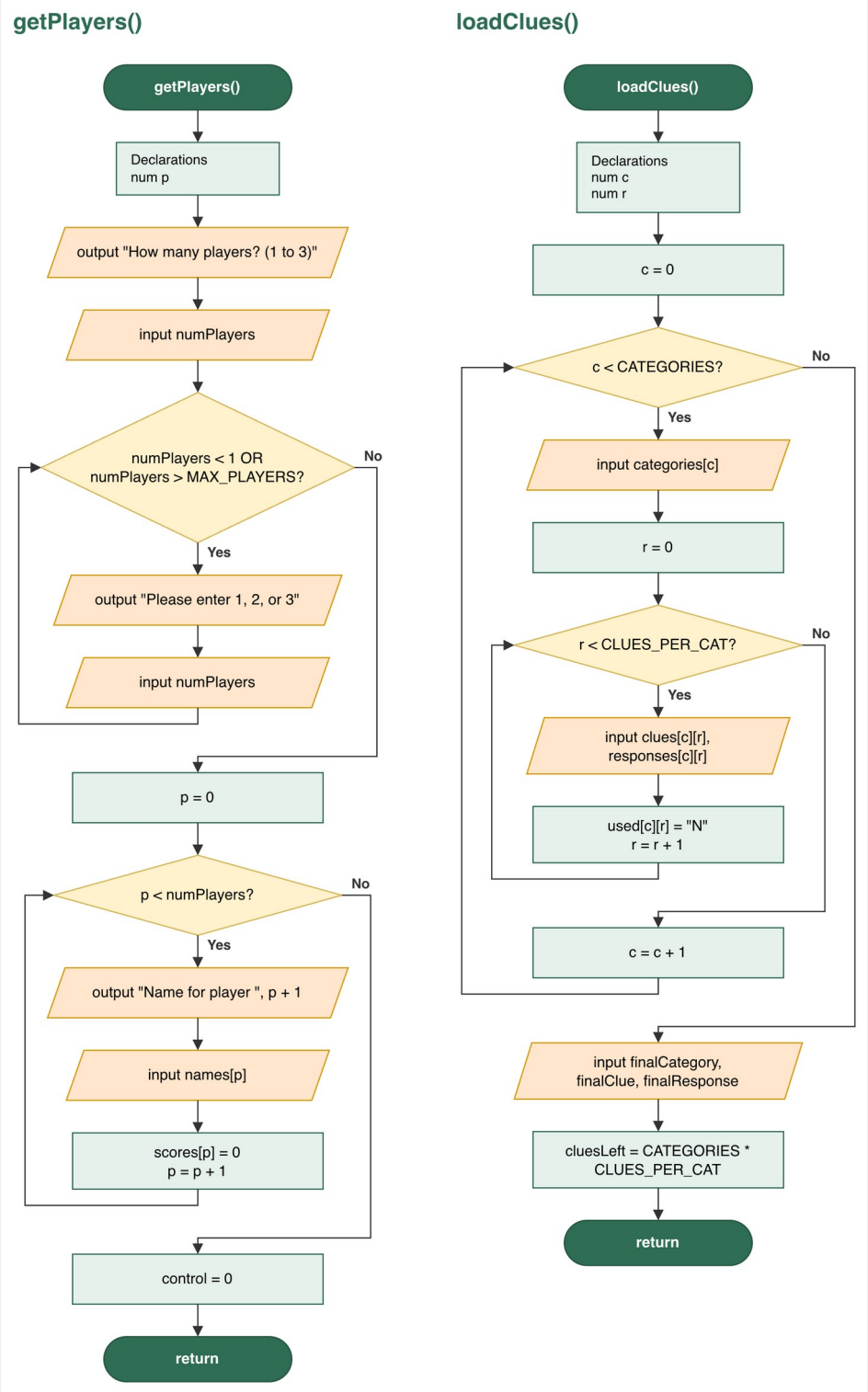


Figure 2 | getPlayers() and loadClues() set up the players and the board

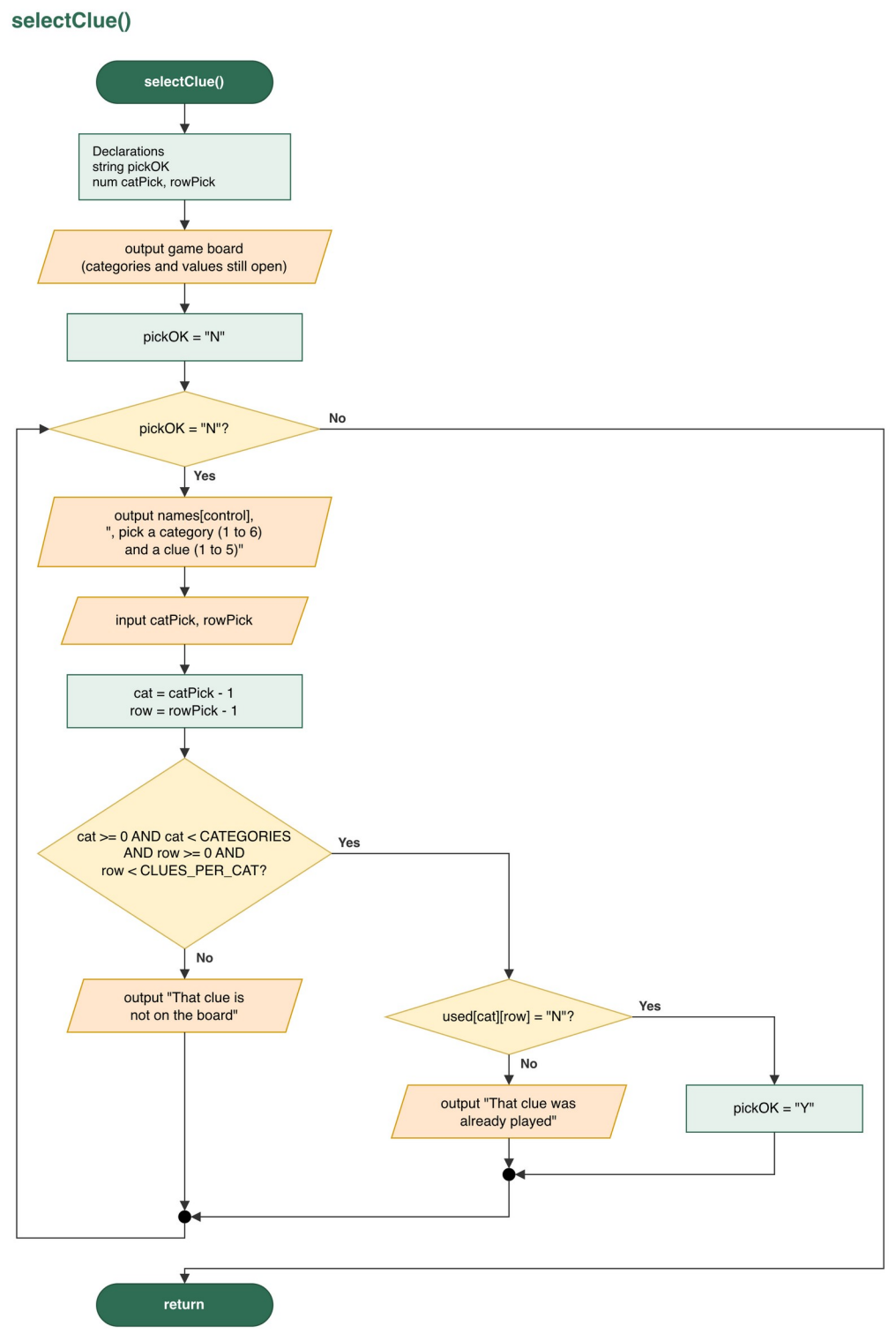


Figure 3 | selectClue() keeps asking until the pick is on the board and has not been played

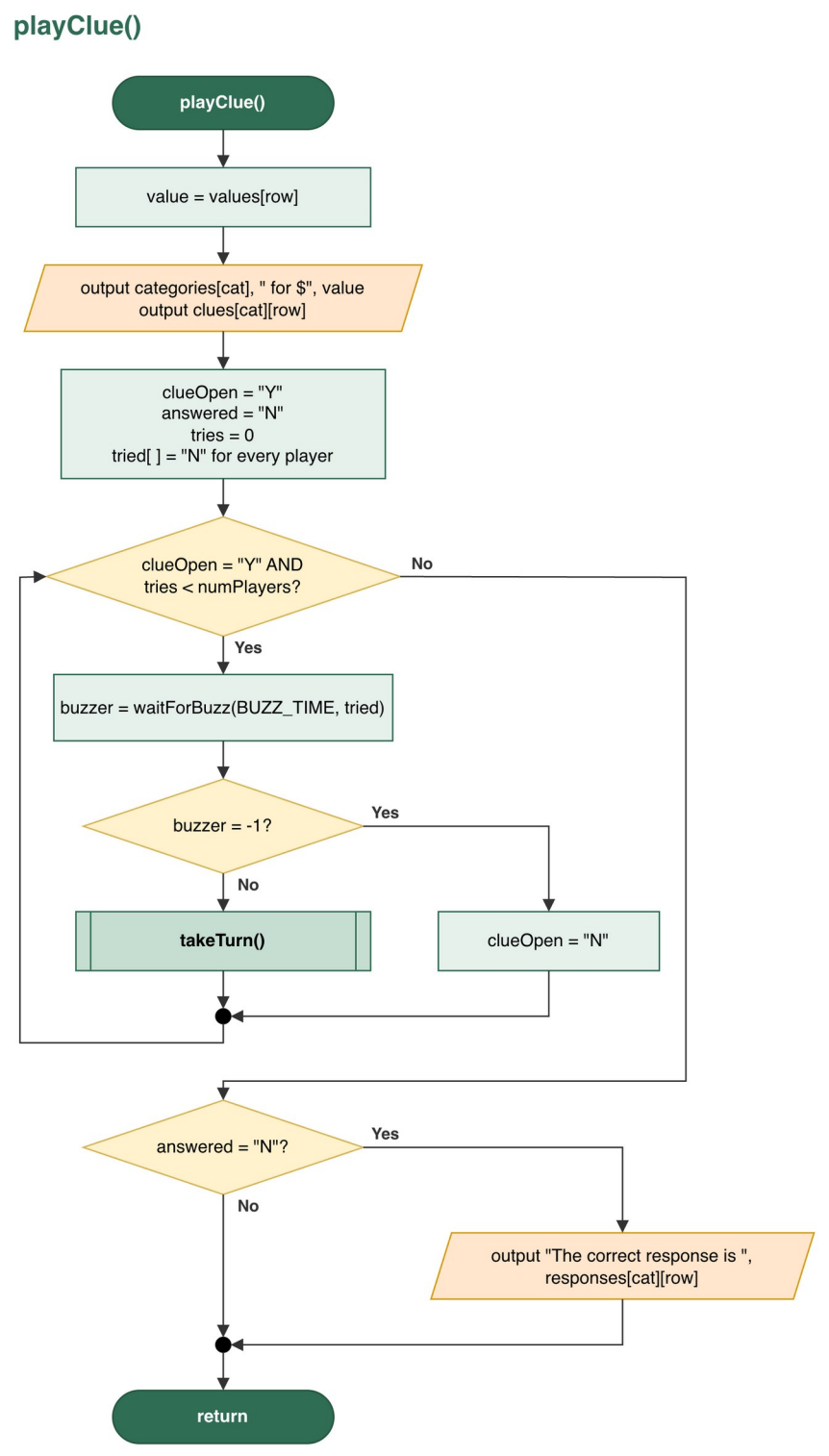


Figure 4 | playClue(). waitForBuzz returns the first player who has not tried yet and buzzes within BUZZ_TIME, or -1 if nobody does

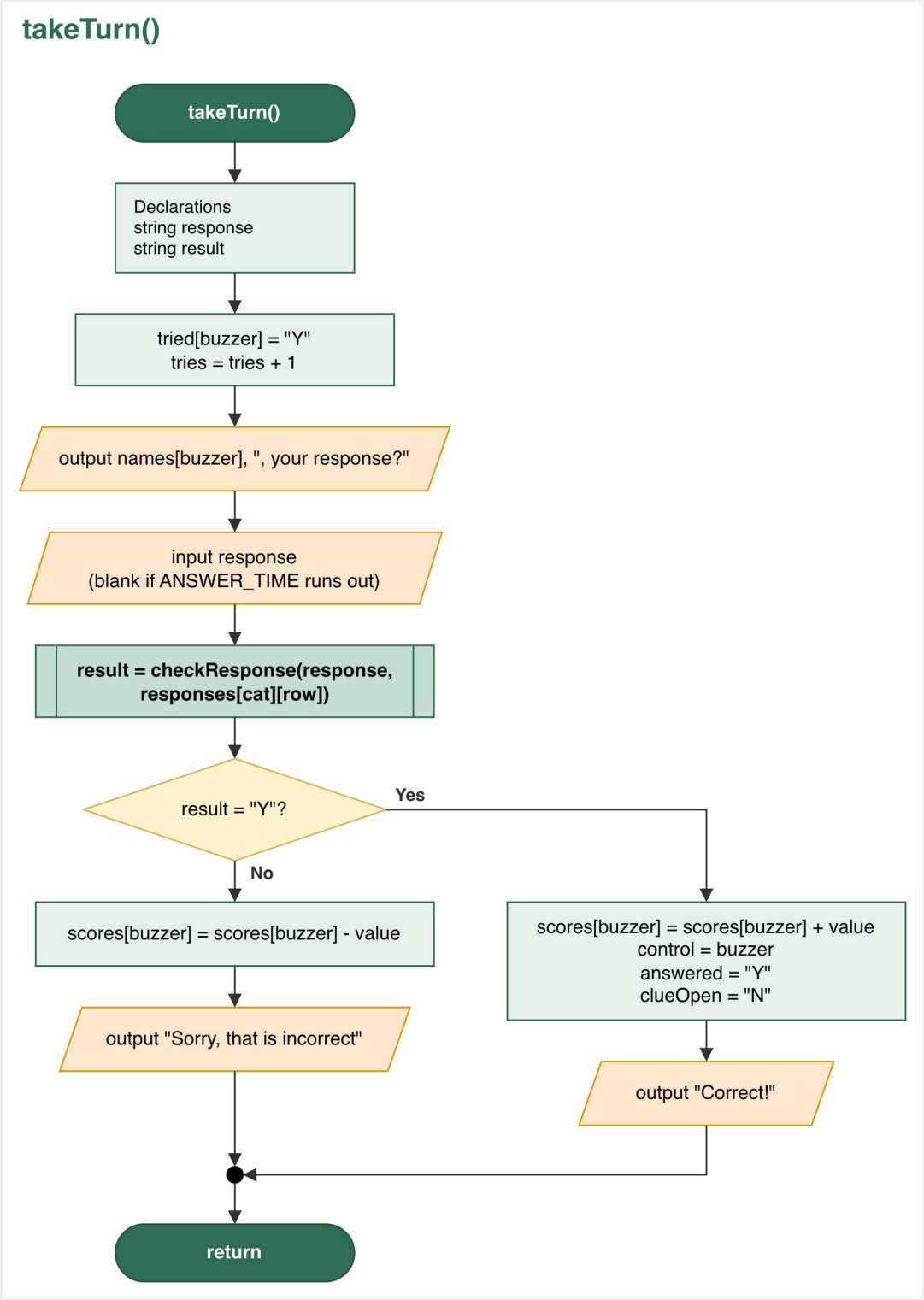


Figure 5 | takeTurn() judges one response and updates the score

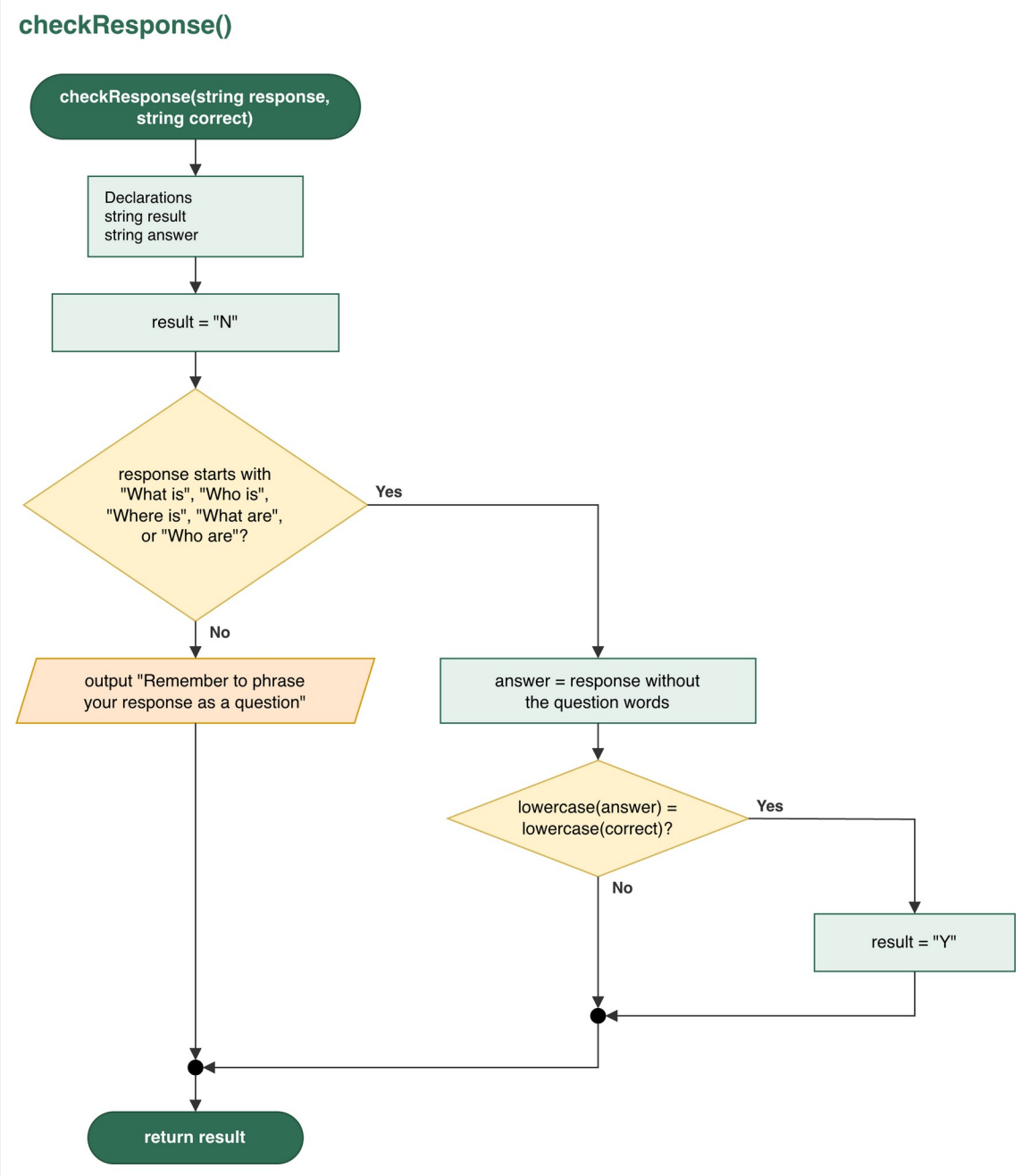


Figure 6 | checkResponse() is shared by takeTurn() and finalJeopardy(), so the logic is only written once

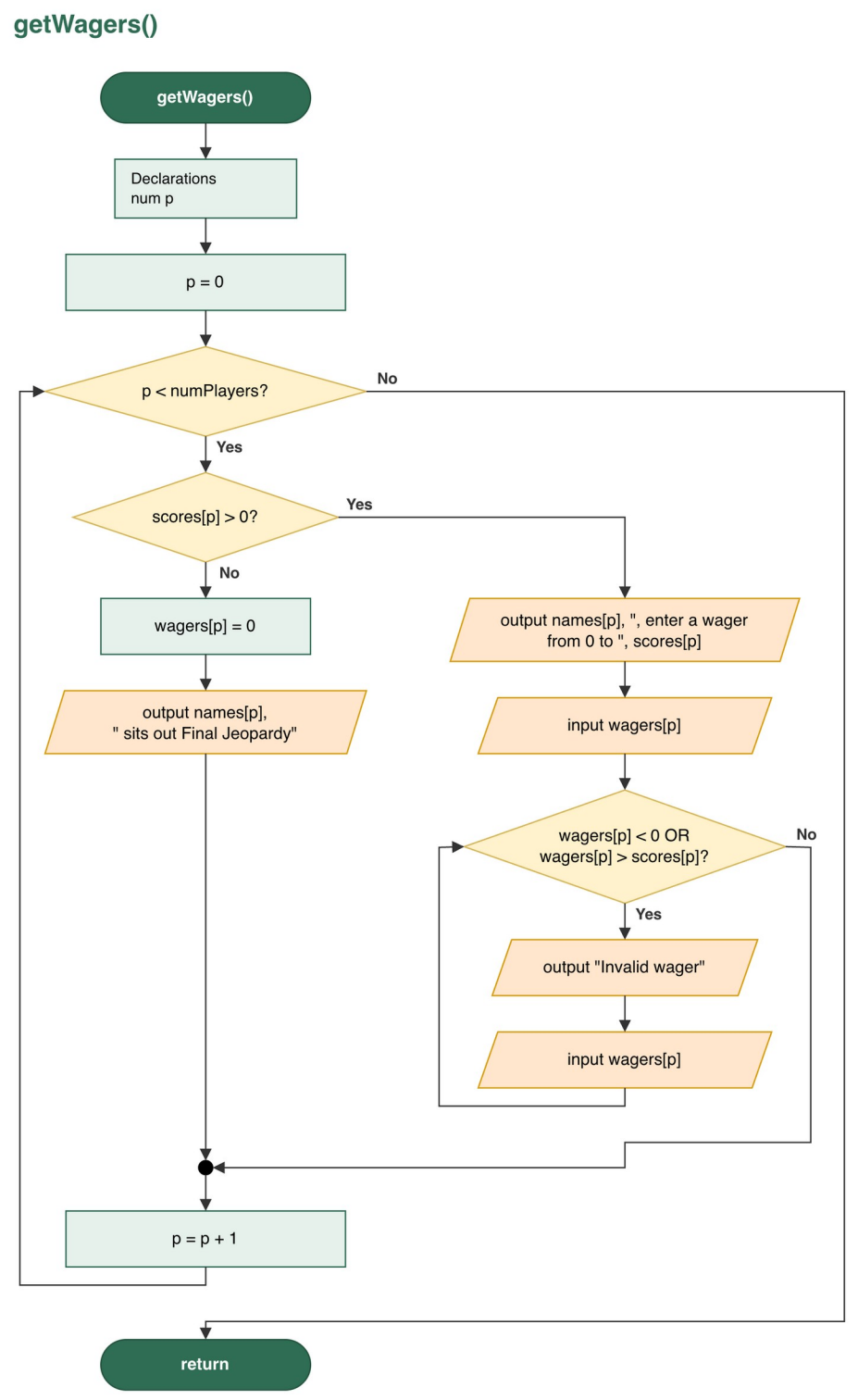


Figure 7 | getWagers() checks that each wager is between $0 and the player's score

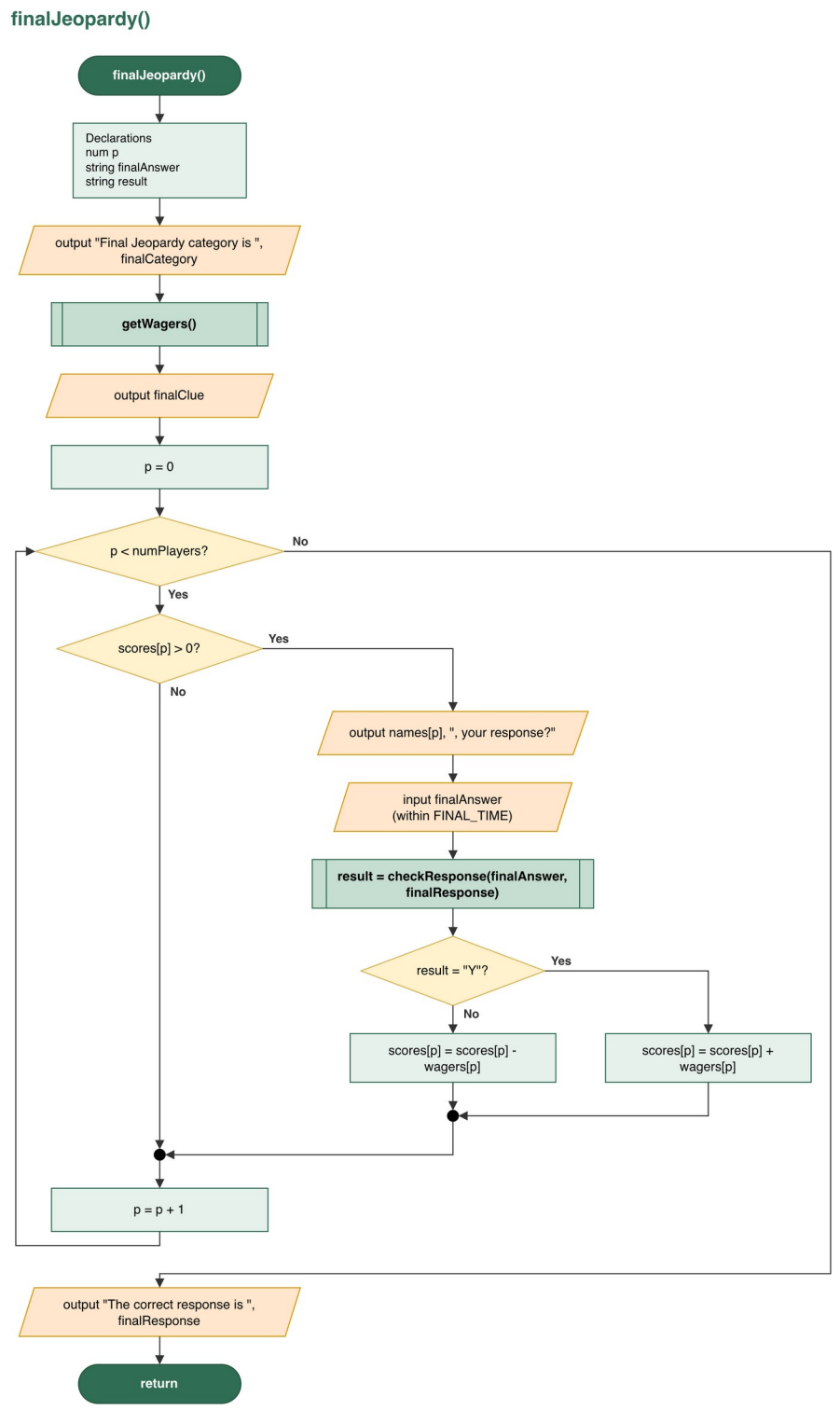


Figure 8 | finalJeopardy() collects each response and applies the wagers

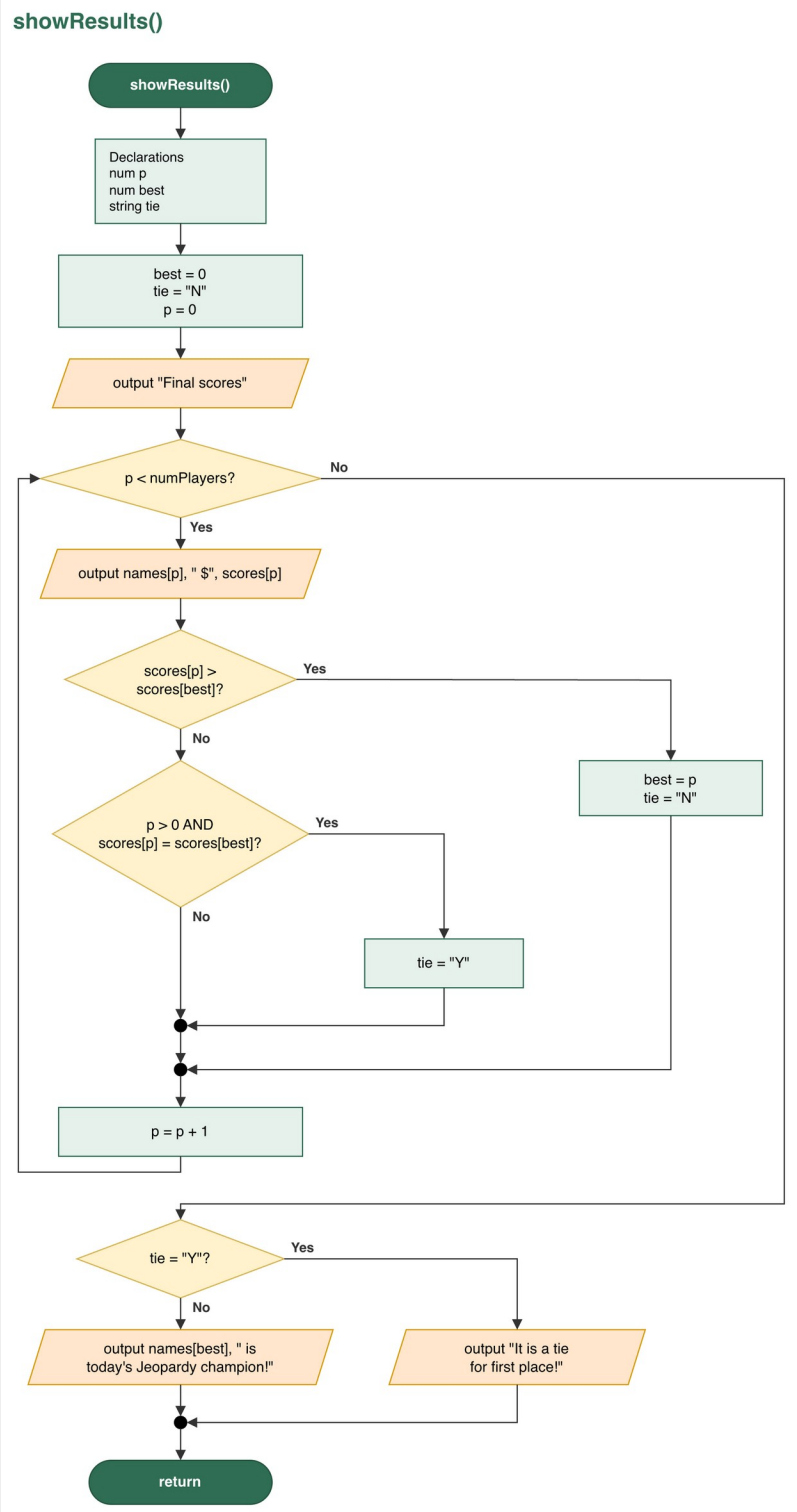


Figure 9 | showResults() finds the champion or flags a tie